JIANG NAN 江南	**Equipment Calibration Policy**	Document No.:	IKQAP#020
		Revision No.	2
	Imperial Kitchen Group Inc.,	Effective Date:	June 28, 2026

1. Purpose

Imperial Kitchen Group Inc. established this policy to ensure the accuracy and precision of all measuring equipment to guarantee consistent, high-quality production. By maintaining strict technical standards, the company upholds its commitment to delivering superior and reliable food products to the community.

2. Scope

This policy mandates that only fully functional and certified measuring equipment be utilized within the stores. All devices must undergo scheduled calibration to ensure they operate within acceptable correction factors, and the use of any equipment with an expired or missing calibration certificate is strictly prohibited. This scope guarantees that every tool used in production meets the required technical standards for accuracy.

3. Responsibilities

3.1. Quality Assurance (QA)

Shall be primarily responsible for verifying and ensuring that all measuring devices remain in good, accurate, and precise working conditions. They handle the official scheduling of equipment calibrations and provide strict oversight for all measurement data documented in the verification logbooks. Additionally, QA tracks continuous equipment performance and upcoming deadlines by documenting them in a Monthly Equipment Performance Report, while maintaining overall accountability for compiling these monthly performance reports.

3.2. Back of House (BOH) Employees

Shall be responsible for the daily verification processes of all operational thermometers with established verification procedures—such as the ice-point or boiling-point methods—once a day. They are also responsible for corrective logging by recording actual readings and applying correction factors to logs if a device exhibits a slight but acceptable deviation. When a device's verification results exceed the allowable tolerance, they perform a measurement adjustment by subtracting the highest recorded error from the equipment's displayed reading to calculate and log the actual measurement. Lastly, they are also tasked to report any equipment defect in the BOH areas, which mandates immediately identifying and removing any measuring device from the production or service area that is damaged, inaccurate, or carries an expired calibration sticker**.**

3.3. Restaurant & Shift Managers

Shall monitor compliance by routinely reviewing daily verification logs to ensure that employees are executing checks correctly and consistently, maintaining overall accountability for logbook monitoring workflows. They manage the store's inventory by keeping a fully updated master list of all measuring equipment on-site, including items like weighing scales, infrared thermometers, and probe thermometers.

3.4. Maintenance / Facilities Management (FM)

Shall be responsible for providing technical support by assisting in the repair or replacement of heavy equipment, such as built-in cold storage thermometers/ gauges, that cannot be calibrated directly by store employees. They are also directly responsible for equipment tagging, which involves ensuring that every individual piece of equipment is clearly labeled with its unique ID number, the date of its last calibration, and its next calibration due date.

3.5. Accredited Third-Party Calibration Agency

Shall be responsible for providing unbiased, technically competent, and legally defensible verification of measuring equipment in strict compliance with the ISO/IEC 17025 standard. They are also responsible for establishing an unbroken chain of metrological traceability to international standards, executing precise calibrations within controlled environments, calculating measurement uncertainty, and recording vital “As-Found” and “As-Left data”.

4. Calibration Validity and Verification

4.1. Calibration Validity/Schedule

This section defines the mandatory timeframe for equipment calibration.

- **New Equipment Procurement:** All newly acquired measuring devices, machinery, or test pieces must be calibrated immediately upon receipt. An exception is made if the unit arrives with a valid calibration certificate, which is recognized for a maximum of one year. However, the calibration date may be moved forward if internal verification procedures indicate that the equipment requires earlier servicing to maintain accuracy.
- **Re-calibration Frequency:** Equipment must be re-calibrated by the date specified by the external calibrating agency. However, this schedule is flexible and may be adjusted based on internal verification results; if a device shows signs of inaccuracy during routine checks, it must undergo re-calibration regardless of the official due date.

4.2. Verification and Correction Factor

To ensure the validity of all measurements, equipment must be verified prior to every use. The verification process is conducted according to the following procedures:

Equipment	Type	Verification procedure	Acceptable Difference	Store Monitoring	Recalibration Frequency
Weighing scale	<5 kgs. capacity	Use of 5 and 50 grams test weight alternatively.	± 2 grams	Daily	**QA:** Quarterly **Third Party:** Every 6 months
	≥5 kgs	Use of 50 grams and 1 kg. test weight alternatively	± 2 grams	Daily	**QA:** Quarterly **Third Party:** Every 6 months
	300 kgs	Use of 1 kg. and 20 kg. (or 50 kg.) test weight alternatively.	± 2 grams	Daily	**QA:** Quarterly **Third Party:** Every 6 months

Equipment	Type	Verification procedure	Acceptable Difference	Store Monitoring	Recalibration Frequency
Thermometer	Probe	**Ice-Point Method (Cold Verification)** 1. Place the thermometer probe directly into the center of the ice bath. 2. Keep the probe submerged for 30 seconds, taking care that it does not touch the container's sides or bottom.	± 2 °C	Daily	**QA:** Quarterly **Third Party:** Every 6 months

Equipment	Type	Verification procedure	Acceptable Difference	Store Monitoring	Recalibration Frequency
		3. Confirm that the device displays a reading of exactly 0°C (32°F). **Boiling-Point Method (Hot Verification)** 1. Place the thermometer probe into the boiling water. 2. Keep the probe submerged for 30 seconds, ensuring it does not make contact with any surface of the container. 3. Confirm that the device displays a reading of exactly 100°C (212°F).			
Thermometer	Infrared	Validated by cross-referencing its readings against data collected from multiple independent infrared thermometers.	± 2 °C	Daily	**QA:** Quarterly **Third Party:** Every 6 months
Freezer/ Chiller Thermometer	Digital	An infrared thermometer will be utilized to measure the unit's digital temperature, allowing for a direct comparison between the external reading and the unit's internal display.	± 2 °C	Twice a week	**QA:** Quarterly **Third Party:** Every 6 months

Equipment	Type	Verification procedure	Acceptable Difference	Store Monitoring	Recalibration Frequency
Room Thermometer	Digital	An infrared thermometer will be utilized to measure the unit's digital temperature, allowing for a direct comparison between the external reading and the unit's internal display.	± 2 °C	Twice a week	**QA:** Quarterly **Third Party:** Every 6 months

Note: If the verification results exceed the allowable tolerance, the actual value must be calculated and logged by
subtracting the maximum recorded error from the equipment's displayed reading:
{Actual Measurement} = {Equipment Reading} -{Highest Error in Reading}

5. Narrative Procedure

Responsibility	Activity
Quality Assurance	1.Verify and ensure that the measuring devices are all under good, accurate and precise conditions 2.Schedules the Equipment Calibration
Store Employees	1.Pre-use Verification: Performing daily "spot checks" or verification procedures (such as the ice-point or boiling-point method) on thermometers before the shift begins. 2.Corrective Logging: Recording actual readings and applying correction factors to logs if a device shows a slight, but acceptable, deviation. 3.Defect Reporting: Immediately identifying and removing any measuring device that is damaged, inaccurate, or has an expired calibration sticker.

Restaurant & Shift Manager	1.Compliance Monitoring: Reviewing daily verification logs to ensure staff are performing checks correctly and consistently. 2.Inventory Management: Maintaining an updated master list of all measuring equipment within the store (e.g., weighing scales, infrared thermometers, and probe thermometers). 3.Schedule Oversight: Ensuring that all equipment is sent for professional calibration before the current certification expires.
Maintenance / Facilities Management (FM)	1.Technical Support: Assisting in the repair or replacement of heavy equipment (like built-in cold storage thermometers) that cannot be calibrated by store staff. 2.Tagging: Ensuring every piece of equipment is clearly labeled with its ID number, date of last calibration, and next due date.
Accredited Third-Party Calibration Agency	1. Professional Calibration: Performs professional external calibration services to validate the accuracy and functional integrity of each device. 2. Official Certification: Issues the official calibration certificates required to validate compliance and track device accuracy history.

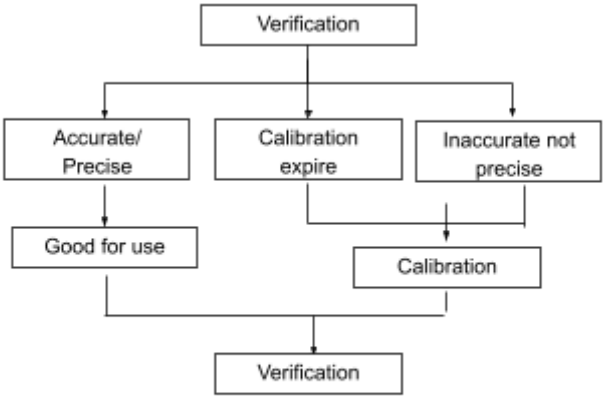
5.1 Summary of Accountability

Action	Responsibility
Daily Verification	Store Employee
Logbook Monitoring	Restaurant/ Shift Manager (OIC)
Monthly Performance Reports	Quality Assurance
Official Certification	Accredited Third-Party Agency

6. Monitoring for Non Conformance and Records

All measurement data must be documented in the verification logbook and subject to oversight by Quality Assurance. Any equipment identified as inaccurate or imprecise during this process must undergo immediate re-calibration; a summary of these verification and calibration workflows is provided in Figure 1. Furthermore, equipment performance and upcoming calibration deadlines are to be documented in a Monthly Equipment Performance Report to ensure continuous tracking.

Figure 1. PROCESS FLOW FOR CALIBRATION



7. Corrective Actions

If a measuring device fails its verification check or exceeds its calibration period, the following Corrective Actions must be implemented to maintain food safety and product standards:

7.1. Immediate Isolation and Identification

When any piece of equipment is identified as inaccurate, structurally damaged, or possessing an expired calibration certificate, it must be decommissioned and removed from the active production or service area immediately to safeguard operational integrity and maintain quality standards. To prevent accidental use by other employees who might be unaware of the hazard, a highly visible "DEFECTIVE" or "OUT OF SERVICE" tag must be securely affixed to the device right away. This labeling serves as a critical communication checkpoint, ensuring the compromised item is easily recognizable as non-operational. Once tagged, the equipment must be transferred to a designated "Hold" or quarantine area, effectively isolating it from functional assets until qualified technicians can perform the necessary repairs, technical evaluations, or formal recalibrations required to restore the instrument to its certified operating specifications.

7.2. Impact Assessment (Traceability)

Upon discovering a malfunctioning device, an immediate evaluation must be conducted on all products measured by that equipment since its last successful verification to assess potential quality deviations. If the evaluation reveals an error significant enough to compromise food safety—such as a walk-in chiller or chest freezer thermometer reading lower than the actual temperature, causing perishable ingredients to be stored in the danger zone where bacteria rapidly multiply—the affected batches must be placed on an immediate hold for thorough testing or ultimately disposed of to mitigate health risks. Finally, the entire scope of this issue, including the evaluation process and the resulting product disposition, must be formally detailed and recorded in a Non-Conformance Report (NCR) to ensure full traceability and regulatory compliance.

7.3. Recalibration or Replacement

When addressing equipment that fails verification, corrective action should first explore internal adjustment; if the device allows for manual correction (such as turning the calibration nut on a dial thermometer), technicians should perform an ice-point or boiling-point adjustment until the unit reads accurately. If internal adjustment is not possible or the device cannot be calibrated in-house, it must be sent to an accredited third-party external recalibration agency equipped to handle complex adjustments. Ultimately, if the equipment cannot be brought back within the acceptable tolerance or corrective factor through either method, it must undergo formal decommissioning, meaning it is permanently discarded and replaced with a brand-new, certified, and calibrated unit to ensure ongoing food safety and operational accuracy.

7.4. Application of Correction Factors

If a device is slightly off but still within a manageable range (and professional calibration is pending), users must apply the Correction Factor to every reading. Example: If a thermometer reads 2 °C higher than the actual temperature, the employee must subtract 2 °C from every recorded measurement to log the Actual Measurement.

7.5. Training and Root Cause Analysis

Following any equipment failure, a formal investigation must be launched immediately to determine the root cause of the malfunction, evaluating whether the instrument was subjected to physical impact such as being dropped, exposed to damaging environmental conditions like extreme heat, or if it has simply reached the natural end of its operational life cycle. Once the underlying cause is identified, targeted refresher training must be conducted for the affected department—whether it be the Commissary, Warehouse, or Store, focusing specifically on the proper handling, transport, and storage protocols required to protect sensitive measuring tools from preventable wear and premature failure.

8. List of Documents

8.1 Equipment Performance Report
8.2 Verification Record/s
8.3 Calibration Certificate/s for each device

9. Change History

Revision No.:	Revision Date:	Author & Position:	Purpose of Revision
0	April 16, 2026	Celine Joy Sanchez**, QA Specialist**	Originator
1	May 25, 2026	Celine Joy Sanchez**, QA Specialist**	• Added Responsibilities
2	June 2, 2026	Celine Joy Sanchez**, QA Specialist**	• Change probe to digital thermometer • Change frequency to Store Monitoring • Added 300 kg for another type of weighing scale • Added recalibration frequency • Added Accredited Third-Party Calibration Agency in Narrative Procedure • Remove QA for In-house Calibration

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